



California Science and Technology University

DevOps with AI Syllabus CSE638, Full 2025

Course & Faculty Information

Lecturer : Qingsong Zhang
E-mail : qingsong.zhang@cstu.edu
Time : Wednesday 7:30pm – 9:00pm (Online)
Saturday 9:00 am - 10:30 am (Onsite)
Contact : 23 Hours
Credit : 1.5 Units
Office hours: By Appointment
Review Date: 11/4/2025

Course Description

DevOps with AI (1.5 units)

This course explores how artificial intelligence (AI) and machine learning (ML) can be integrated into DevOps practices to automate, optimize, and enhance software development and operations. Students will gain hands-on experience with AI-assisted CI/CD pipelines, predictive analytics, intelligent monitoring, automated remediation, and security.

By the end of the course, students will be able to:

1. Understand how AI and ML can augment DevOps processes.
2. Implement AI-driven automation in CI/CD pipelines.
3. Use predictive analytics for reliability and performance improvements.
4. Apply AI for intelligent monitoring, anomaly detection, and incident response.
5. Evaluate AI tools and frameworks for DevOps at scale.

Textbook and Reference Material

Textbook Information

Mastering ChatGPT and Prompt engineering, ISBN-13: 979-8390306932, ASIN: B0C1J1H9WQ

Amazon Item: ([Unlock the Full Potential of AI Language Models: Power of ChatGPT for Success](#))

Reference Information

- 1). **OpenAI** (<https://platform.openai.com/docs/introduction>)

Prerequisite

Working experience with Python Programming, basic computer concepts, business software tools, such as Anaconda.

Instructional Method

A combination of lectures, class participation, and hands-on work are used to facilitate the learning. Projects will be assigned individually or as a group. Homework includes, but is not limited to course readings, assignments, and courses projects. Students are encouraged to seek instructor assistance, engage in group discussion, and ask for advice from the school advisor.

Out of Class Work

Students are expected to engage in a minimum of two hours of out-of-class work for every hour of classroom instruction. Out-of-class work includes, but is not limited to course readings, group projects, working on course problems or case studies, exam preparation, course research, and group discussions. Out-of-class work is not simply an exercise for students to complete while not in class. Out-of-class work helps a student solidify the learning and helps to prepare the student for learning the next lecture. Out-of-class work will be assessed and graded and will impact the student's final grade.

Learning Resource Center

Students will be required to use the library for study. The librarian or library assistant is available to students to provide assistance as well as guidance on navigating the online resources of the Learning Resource Center. CSTU website has provide details instructions on how to access the Learning Resource Center and search for articles in a specific field.

OBJECTIVES

To accomplish its mission, CSTU is committed to enhance student competencies by:

- Providing working adults with higher educational opportunities that are flexible and accessible.
- Providing graduate level students professional training and preparation for careers
- Providing higher educational opportunities that are current with technology and career demands.
- Providing faculty members that have demonstrated expertise in, their respective domain, both professionally and academically.
- Integrating into the educational process a better understanding of cultural diversity needs.
- Delivering educational support services that meet student life demands and schedules.

- Building within students a value for life-long learning and education.
- Providing educational resources in a manner that effectively uses current technology.

CSTU is committed to the highest ethical standards in the pursuit of the mission. The policies, procedures, and standards guide CSTU core values set forth below. These values are honored in our daily structure and activity as members of this community. We are committed to respecting the rights and dignity of others while conducting ourselves with integrity in our dealings with and on behalf of all individuals in our environment and are accountable as individuals and as members of this community for the ethical conduct and for compliance with applicable laws, University policies, and directives while conscientiously strive for excellence in our work.

Student Learning Outcomes

The University Student Learning Outcomes assessed and reinforced in this course include but are not limited to the following:

- Communication
- Critical Thinking
- Information Literacy
- ML and AI fundamentals

Attendance Policy

The University recognizes the correlation between attendance and both student retention and achievement. Per University Policy **Students are expected to attend all classes, actively participate and complete all assigned course work for all courses for which they are registered.**

Evaluation

Assessment may be determined by participation, out of class work, quizzes, tests, projects, presentation of the project,

Evaluation Criteria & Grading Scale

Grading Scale:

*** The following grading scale will be used:*

PERCENT	GRADE	PERCENT	GRADE		Mid-term Exam... 15%
98%-100%	A+	78%-79%	C+		Group Project..... 20%
93%-97%	A	73%-77%	C		Case Discussions 30%
90%-92%	A-	70%-72%	C-		Final Exam..... 20%
88%-89%	B+	68%-69%	D+		Class Participation <u>15%</u>
83%-87%	B	63%-67%	D		Total 100%
80%-82%	B-	60%-62%	D-		
		Below 60%	F		

WEEKLY SCHEDULE

Week 1: Foundations of AI-Assisted DevOps

Session 1: Introduction to DevOps and AI Convergence

1. Evolution of DevOps and the need for AI integration
2. Key challenges in modern DevOps environments
3. AI vs traditional automation in DevOps
4. AIOps and AI-assisted DevOps use cases
5. Industry trends and case studies

Session 2: AI/ML Techniques in DevOps

1. Overview of ML categories: supervised, unsupervised, reinforcement learning
2. Data sources in DevOps (logs, metrics, traces, configs)
3. AI algorithms for prediction and anomaly detection
4. DevOps data pipelines and feature engineering
5. How AI models integrate with DevOps toolchains

Lab (Week 1):

1. Set up a cloud-based DevOps lab (AWS/Azure/GCP)
2. Collect sample CI/CD and system monitoring data for AI tasks

Assignment:

- Write a short report on 3 real-world applications of AI in DevOps and evaluate their impact.

Week 2: AI Tooling & Cloud Platforms for DevOps

Session 3: AI-Powered DevOps Tools

1. GitHub Copilot, AWS CodeGuru, Azure DevOps AI features
2. Comparison of AIOps tools: Datadog, New Relic, Dynatrace
3. Capabilities: anomaly detection, prediction, optimization

4. Strengths and weaknesses of vendor platforms
5. Building toolchains with AI plug-ins

Session 4: Setting up AI-DevOps Environments

1. Using cloud AI services (SageMaker, Vertex AI, Azure ML)
2. Configuring CI/CD pipelines with AI-assisted tools
3. Secure secrets and credential management
4. Managing data for ML workflows
5. Lab walkthrough: building a basic AI-augmented pipeline

Lab (Week 2):

1. Deploy a simple Jenkins or GitHub Actions pipeline with AI-based code quality analysis

Assignment:

- Compare at least two AI-augmented DevOps tools and propose an integration plan for a mid-sized software team.

Week 3: Intelligent CI/CD Pipelines

Session 5: AI in Code Quality and Testing

1. Automated static code analysis using ML
2. AI-driven test generation and prioritization
3. Reducing flaky tests with ML
4. Measuring test coverage and quality with AI
5. Tools and frameworks for AI-enhanced QA

Session 6: Pipeline Optimization

1. Using ML to reduce pipeline execution time
2. Dynamic build resource allocation
3. Predictive build failure detection
4. Caching and artifact management with AI insights
5. Continuous feedback loops

Lab (Week 3):

1. Implement an AI-based test prioritization plugin in a CI pipeline

Assignment:

- Build a CI pipeline that uses AI predictions to skip unnecessary builds and tests.

Week 4: Predictive Analytics in DevOps

Session 7: Risk Prediction in Deployments

1. Predicting failed deployments using historical data
2. Rollback strategy recommendations
3. ML models for deployment success scoring
4. Data labeling challenges
5. Case studies from industry

Session 8: Capacity and Performance Forecasting

1. Forecasting infrastructure usage
2. Identifying bottlenecks using AI models
3. Capacity scaling policies driven by ML

4. Performance prediction for microservices
5. Leveraging time-series forecasting algorithms

Lab (Week 4):

1. Build and train a time-series forecasting model to predict CPU and memory usage

Assignment:

- Create a report analyzing the effectiveness of AI forecasting for autoscaling in Kubernetes clusters.

Week 5: Intelligent Monitoring and Observability

Session 9: AI-Driven Anomaly Detection

1. Using ML on logs and metrics for anomalies
2. Unsupervised learning approaches (clustering, isolation forests)
3. Real-time vs batch detection
4. Noise reduction and false positives
5. Best practices for integrating detection in pipelines

Session 10: Root Cause Analysis with AI

1. Graph-based algorithms for dependency mapping
2. Log correlation using NLP
3. AI-driven alert grouping
4. Prioritizing incidents based on impact
5. Automating RCA reporting

Lab (Week 5):

1. Implement log anomaly detection using unsupervised learning on sample logs

Assignment:

- Build a system that identifies anomalies in logs and sends AI-prioritized alerts.

Week 6: Automated Incident Response and Remediation

Session 11: Self-Healing Systems

1. What is self-healing architecture?
2. Automated rollback and failover using ML triggers
3. Decision trees and reinforcement learning for response
4. Orchestration with Kubernetes Operators
5. AI as a “virtual SRE”

Session 12: Alert Prioritization and Triage

1. Intelligent alert suppression
2. Correlation-based incident grouping
3. AI playbooks for triage
4. Integration with ITSM tools (PagerDuty, ServiceNow)
5. Tracking remediation effectiveness

Lab (Week 6):

1. Create a Python-based AI bot that triggers automated remediation on a simulated failure

Assignment:

- Implement a self-healing mechanism for a microservice using AI-driven triggers.

Week 7: Generative AI for DevOps

Session 13: AI for Configurations and IaC

1. Generating Terraform/CloudFormation templates using LLMs
2. Policy-as-code automation
3. Avoiding prompt injection and errors
4. Building reusable automation modules
5. Hands-on with GPT-based code assistants

Academic Integrity

CSTU expects that all academic work submitted by students be original, or in the case of cited material, properly acknowledged as the work, ideas, or language of another. Further, all acts of academic dishonesty are strictly prohibited. These include, but are not limited to, cheating, plagiarism, fabrication, unauthorized collaboration, misappropriation of resource material, or any other violation of University regulations.